GLOBAL SCHOOL OF SCIENCE

Second Terminal Examination - 2082

Class: XI **Subject: Mathematics (0071)** **Full Marks: 75**

Time: 3 hrs. **SET 'B'** **Pass Marks: 30**

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. **[11 × 1 = 11]**

1. The number of proper subsets of the set $A = \{a, e, i, o, u\}$ is

   a) 25 b) 32 c) 31 d) 15

2. If $A = (2, 7]$ and $B = [4, 9)$, then $A \cap B$ is

   a) $[4, 7]$ b) $(4, 7]$
   c) $[4, 7)$ d) $(4, 7)$

3. The domain of the real valued function $f(x) = \frac{1}{x-2}$ is

   a) $\mathbb{R} - \{-2\}$ b) $\mathbb{R} - \{2\}$
   c) $\mathbb{R} - \{1\}$ d) $(-\infty, \infty)$

4. The logarithmic function$y = \log_a x$, is defined if

   a) $a > 0$ b) $a < 0$
   c) $a \neq 1$ d) both (a) and (c)

5. If α and β be two roots of the quadratic equation $x^2 - 2x - 3 = 0$ then $\alpha^2 + \beta^2 =$

   a) 11 b) 12 c) 9 d) 10

6. The value of $\sqrt{-4} \times \sqrt{-9}$ is

   a) -6 b) $-6i$ c) $6i$ d) 6

7. The lines represented by $ax^2 + 2hxy + by^2 = 0$ will be perpendicular if

a) $h^2 = ab$ b) $h^2 = 2ab$

c) $a + b = 0$ d) $a = b$

8. The perpendicular distance of the point $(-1, 2)$ from the line $3x - 4y + 1 = 0$ is

a) 5 b) 2 c) 3 d) 4

9. $\lim\limits_{x \to 1^-} \frac{x-1}{|x-1|} =$

a) 1 b) -1 c) 0 d) does not exist

10. The derivative of $\sin x$ with respect to $\cos x$ is

a) $\cot x$ b) $-\tan x$ c) $\tan x$ d) $-\cot x$

11. The derivative of $\cos^{-1} x$ with respect to x is

a) $\frac{-1}{\sqrt{1-x^2}}$ b) $\frac{1}{\sqrt{1-x^2}}$

c) $\frac{1}{\sqrt{1+x^2}}$ d) $\frac{-1}{\sqrt{1+x^2}}$

Group 'B'

Short Answer Questions. **[8 × 5 = 40]**

12. a) Construct the truth table of $q \wedge (\sim p \wedge q)$. Also, comment on the result. [1+2]

b) Solve the inequality: $|x - 1| > 1$. [2]

13. a) Let the function $f: A \to B$ be defined by $f(x) = \frac{x+3}{x-1}$ with $A = \{-3, -2, -1, 0, 2, 3\}$ and $B = \{-3, -1, -\frac{1}{3}, 0, 1, 3, 5\}$. Find the range of f. Is the function f one to one and onto both? If not, how can the function be made one to one and onto both? [1+1+1]

b) If $a^2 + b^2 = 2ab$, then prove that : $\ln\left(\frac{a+b}{2}\right) = \frac{1}{2}(\ln a + \ln b)$. [2]

14. a) Find the domain and range of the real valued function $f(x) = \sqrt{x^2 - 4x - 5}$. [3]

b) Prove that : $(1 + i)^4 \left(1 + \frac{1}{i}\right)^4 = 16$. [2]

15. a) Find all possible values of z if $z^2 = 12 - 5i$. [3]

b) If α and β are the roots of the equation $ax^2 + bx + b = 0$, then show that: $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{b}{a}} = 0$. [2]

16. a) Let the equations of two straight lines be $18x + y - 5 = 0$ and $3x - 2y + 1 = 0$.

i) Find the angle bisectors between them. [1]

ii) Show that bisectors are perpendicular to each other. [1]

iii) Also, identify the acute angle bisector. [1]

b) If p be the length of perpendicular from the origin on the line $\frac{x}{c} + \frac{y}{d} = 1$, then prove : $\frac{1}{c^2} + \frac{1}{d^2} = \frac{1}{p^2}$. [2]

17. a) Evaluate : $\lim_{x \to \theta} \frac{x\cos\theta - \theta\cos x}{x - \theta}$ [3]

b) If the equation $(1 + m^2)x^2 + 2mcx + c^2 - a^2 = 0$ has equal roots, show that: $c^2 = a^2(1 + m^2)$. [2]

18. a) From the first principle, find the derivative of $\sqrt{3x - 5}$. [3]

b) If $x = a(1 + \cos t)$ and $y = a(t - \sin t)$, then prove that $\frac{dy}{dx} = tan\frac{t}{2}$ [2]

19. a) Write the derivative of $cosecx$ and $tanx$ with respect to x. [1]

b) Find $\frac{dy}{dx}$ if $x^2 + y^2 = \cos(xy)$. [2]

c) Evaluate: $\lim_{x \to \infty} 2x\left[\sqrt{x^2 - 1} - x\right]$ [2]

Group 'C'

Long Answer Questions. **[3 × 8 = 24]**

20. a) Let A, B and C be three non-empty sets. Define $B - A$ in set builder form. Also, prove that: $A - (B \cap C) = (A - B) \cup (A - C)$ [1+2]

b) Let $f: \mathbb{R} - \{3\} \rightarrow \mathbb{R} - \{1\}$ be a function defined by $f(x) = \frac{x}{x-3}$. Show that f is bijective. Also, find $f^{-1}(x)$. [2+1]

c) Solve the inequality : $|4x + 2| \leq 6$ and show the result in real number line. [1+1]

21. a) Write the condition that the quadratic equations $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ may have both roots common. Find the values of m and n so that the equations $mx^2 - 5x + n = 0$ and $5x^2 - mx + 3 = 0$ have both roots common. [1+2]

b) Write the condition for a general equation of second degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ to represent a pair of lines. Find the value 'k' for which the equation $x^2 + kxy + y^2 - 5x - 7y + 6 = 0$ represents a pair of lines. [1+2]

c) If the distance between two parallel lines $3x - 4y + 5 = 0$ and $6x - 8y + k = 0$ is $\frac{3}{10}$ units, then find the value(s) of k. [2]

22. a) When does a function $f(x)$ become continuous at any point $x = a$? [1]

b) Let a function $f(x)$ be defined by

$$f(x) = \begin{cases} 2x - 3 & for \quad x > 3 \\ 4 & for \quad x = 3 \\ x^2 - 6 & for \quad x < 3 \end{cases}$$

i) Verify that the limit of function exists at $x = 3$. [1]

ii) Is function continuous at $x = 3$? Give reason. [1]

iii) If $f(x)$ is not continuous, mention the type of discontinuity. [1]

iv) Redefine the function $f(x)$ to make it continuous at $x = 3$. [1]

c) Find $\frac{dy}{dx}$ if $x^3 - y^3 + 3bxy = 0$. [3]

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